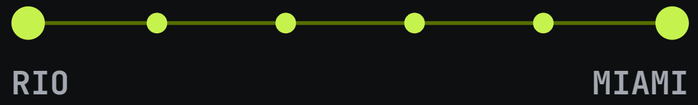
TESTNET · TEST KEYS · NOT REAL CUSTODIANS

Radox Custody Rail

One record for the goods, from origin to delivery, that every company along the way could share.

Scan: the live shipment
rio-miami-rail.vercel.app

Toward a SWIFT-like standard on Stellar · Pilot corridor: Rio–Miami



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PMF · THE PROBLEM

Who holds the whole chain?

Warehouse — Port — Carrier — Customs — Recipient

Our hypothesis: each company records only its own leg; no shared record follows the goods. Not yet tested with a customer.

45M

ocean bills of lading issued a year

DCSA, Feb 2023 (2021 data)

~11%

of bills of lading issued electronically, 2025

DCSA, as of 2025 (Jan 2025 guide: 5.7%)

22%

of recorded cargo thefts in 2025 involved insiders

TT Club & BSI Cargo Theft Report, Apr 2026

Context: our explainer video about a Rio–Miami trade-corridor initiative poses one possibility: a unique digital record for every shipment that both sides can check. This project is a custody layer for that record.

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PMF · WHAT IT COSTS

Cargo goes missing. A bank got fooled.

Our hypothesis: a financier cannot see who holds the goods before paying. No customer has confirmed it yet.

R$900M

est. direct loss, 8,570 cargo robberies in Brazil, 2025

NTC&Logística survey, 28 Mar 2026

US$85M

HSBC's loss on forged documents for oil trades that never happened

Hin Leong case, Global Trade Review, 18 Nov 2024

US$2.5T

global trade finance gap (market size only)

ADB survey of banks, Dec 2025

These numbers size the market problem. They are not results of ours, and we do not claim our build prevents any of it.

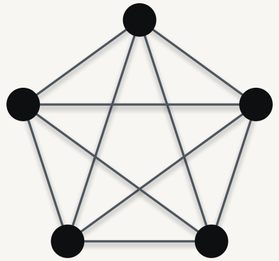
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THE PROPOSAL · A PUBLIC STANDARD

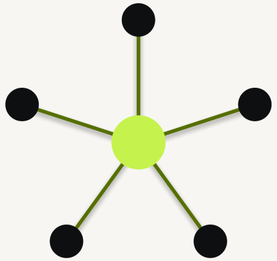
A SWIFT-like standard: integrate once

The goal is a shared, SWIFT-like custody standard that companies along a supply chain, from origin to destination, could use. **Neither the standard nor the rail exists yet.**

PAIR BY PAIR · 10 LINKS



ONE STANDARD · 5 LINKS



Example, 5 companies: 10 custom links, or 5 links to one standard.

OPEN · ANYONE

— Record format, roles and events: the custody_registry contract (Apache-2.0)

— Verify any shipment, no login (slide 6)

RADOX OPERATES · CLIENT PAYS

— Onboarding, issuance controls, operation of the rail

— The API and backend: closed, not self-hostable

— Subscription per connected institution + fee per shipment; no prices yet (slide 9)

Each company would integrate once instead of pair by pair, but would still need a key and one connection. No port, airport or chamber link: none confirmed. Radox is the rail, never the issuer. Regulation and licences stay with the client and its local counsel.

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THE RAIL · DESIGN, LIFECYCLE BUILT

One rail, origin to delivery

One record would follow the goods across company boundaries; today it runs across five test keys.



— **Rule of every link:** both sides sign one Stellar transaction (delivery: recipient and holder), or the call fails.
— Each step extends the on-chain `head`: anyone can replay the history from Horizon and check it against the head.
— The token stays with its holder in transit; only custody moves.

Built: this lifecycle on testnet, test keys. Design, not built: real companies on the rail, the standard, integrations.

Keyed by sha256(container | origin); would sit beside the carrier's bill of lading, not integrated. [1] Radox policy, not enforced on-chain; redeem needs the holder's signature.

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THE PROOF · LIVE ON TESTNET

The lifecycle ran on testnet

Warehouse → Port, both signed	ledger 5381142 · 635c99…f0bd38
Port → Carrier, both signed	ledger 5381143 · 56a0f0…7b7cc2
Event attested by Source	ledger 5381144 · 333792…068905
Carrier → Customs, both signed	ledger 5381145 · ac268a…c3c4b2
Customs → Recipient, both signed	ledger 5381146 · 5199d4…400b7a
Delivered by Recipient to Holder, both signed	ledger 5381147 · 919def…144e37
Redeemed: token burned	864fd1…15644d

Verify head

`2b3d51f0` open · operator
`635c99ac` handover 1 · Warehouse → Port
`56a0f006` handover 2 · Port → Carrier
`ac268a49` handover 3 · Carrier → Customs
`5199d4fb` handover 4 · Customs → Recipient
`919def8c` deliver · Recipient → Holder
`864fd164` burn · existing redeem

7 successful

9 refused, e.g. wrong signer (simulated)

24 unit tests

MATCH head check, 25 Oct

`custody_registry`
`CCBJVXMJYWPHTDJHDFPRGKPGXGEP24JOPJAMUS5C3BMZJTRTWR2H4LII`

Second shipment (proofs/three-machine.json): two team members each signed their handovers from their own computers and keys; still test keys.

All test keys: this proves the mechanism, not independent companies.

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ECOSYSTEM · WHY STELLAR

Why Stellar: all signers, one call

01	**Multi-party auth in one call**	Soroban authorises both companies of a link inside one contract invocation: the handover is atomic.
02	**History from Horizon, checked by an on-chain head**	RPC keeps contract events about 7 days, so we replay each shipment's transactions from Horizon and check them against the head stored in the contract.
03	**The token stays put, then burns**	The shipment token never moves in transit; the existing Radox redeem burns it after delivery.

Nearest neighbours we found on Stellar (LumenLoop, SCF, Scout, 24–25 Oct): Kutana and Trustless Work release escrow on trade milestones; Pipeline finances commodity trade. We found none that records a custody handover signed by two custodians, and no SEP.

Open source, Apache-2.0: a custody primitive other Stellar real-world-asset issuers can reuse.

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HACKATHON PROGRESS

Built this weekend, on disclosed prior work

PRIOR WORK · BEFORE SUN 25 OCT 09:00

— Radox platform (closed backend); its 3 Soroban contracts are on Stellar mainnet, public since 2026-08-19

— transfer_agent and its redeem (new testnet instance from the existing wasm)

— Architecture notes (20 Oct), planning files (24–25 Oct), the explainer video, our agent tooling (from 22 Oct)

— No contract code before then: first repo commit 25 Oct 09:00 UTC

BUILT AT HACKMERIDIAN

— custody_registry contract, 24 unit tests

— Lifecycle harness: 7 testnet transactions, 9 refused calls

— verify-head script, the public shipment page, this deck

— Radox API reads the registry: same head (closed backend, testnet)

AI attribution Code, tests, scripts, the shipment page, docs and this deck were made with AI coding agents (Claude Code); the team decided scope and what ships. Per-file record in AI_USAGE.md. The video narration is an AI voice generated from Pedro's voice.

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MARKET · MODEL · THE ASK

One rail, every corridor

MARKET · FIGURES, NOT OURS

US$94.2B

US–Brazil goods trade, 2025[1]

US$6.1B

Florida's goods exports to Brazil, its #1 market, 2025[2]

Global: US$26.26T world merchandise exports, 2025[3]; ~80% of goods trade by volume goes by sea[4]; ~45M ocean bills of lading a year[5].

MODEL · NO PRICES YET

Subscription per connected institution + a fee per shipment.

Borrows the shape of Swift's billing: annual charges, plus traffic fees for messages sent[6]. Not affiliated with Swift.

Now testnet only; no users or revenue yet. **Next** one Rio–Miami pilot (none agreed yet), SCF Build, audit, mainnet.

[1] US Census Bureau, Trade in Goods with Brazil, 2025 (exports + imports, sum ours) [2] USTR, Florida trade data [3] WTO, Mar 2026 [4] UNCTAD, Review of Maritime Transport [5] DCSA, Feb 2023 [6] swift.com billing rules. Not Radox volumes.

Our ask **Introductions to one financial institution and one port, airport or logistics operator for a first pilot shipment, and a Stellar engineer's review of the design.**

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THE TEAM

Why us: a founder already on Stellar mainnet

Pedro Wakigawa Saragossy

Founder, captain. Built Radox: 3 Soroban contracts on Stellar mainnet.

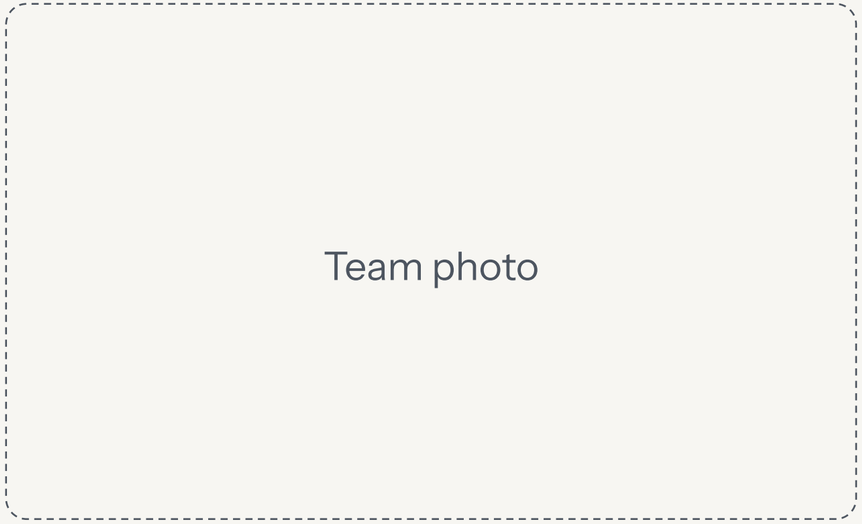
Leo Schlanger

Developer: custody_registry, the testnet harness and the proofs.

Vhendala

QA and product review: reads every deliverable as a judge would.

Leo and Vhendala joined for the hackathon.



Rio–Miami first. Our goal: one rail for goods, every corridor.

Live page rio-miami-rail.vercel.app **Code** github.com/pwsaragossy/rio-miami-rail (Apache-2.0)

AI Code, page, docs and deck made with AI coding agents (Claude Code). Narration: an AI voice from Pedro's voice. No AI-generated images in this deck.

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